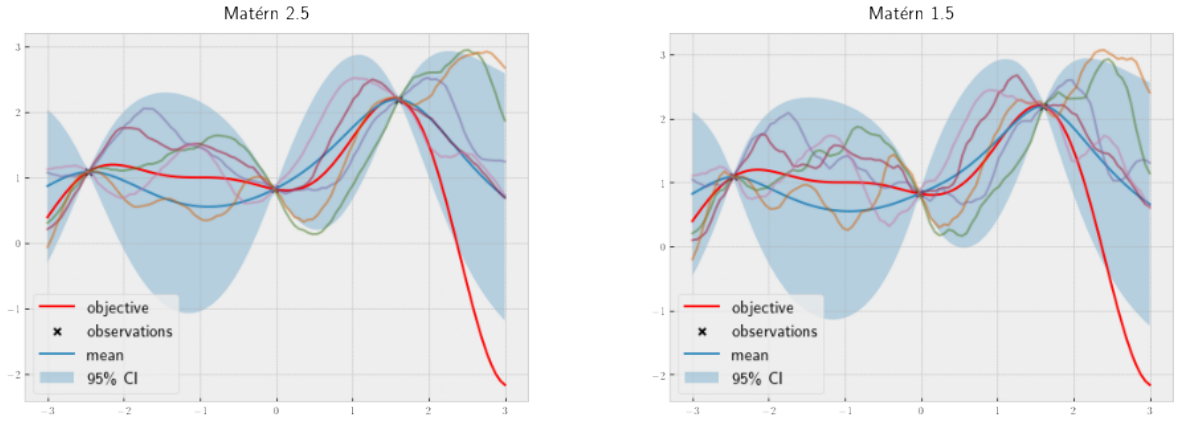


Figure 3.14 GP's predictions with the Matérn 2.5 (left) and Matérn 1.5 (right) kernel. The samples here are smooth enough for the GP to effectively learn from data, but are also jagged enough to realistically model real-life processes.

We see that the samples from this 2.5 kernel is much smoother, which leads to more effective learning by the GP. However, these samples are also rough enough that they resemble functions that we might see in the real world. For this reason, most efforts, both research and applied, in Bayesian optimization utilize this Matérn 2.5 kernel. In future chapters when we discuss decision-making for Bayesian optimization, we will accordingly default to this kernel.

> **NOTE** **While we are not including the corresponding details here, a Matérn kernel has its own length scale and output scale, which may be specified to further customize the behavior of the resulting GP in the same manner as in the previous subsection.**

By pairing a mean function with a kernel, we can induce complex behavior in the predictions of the GP. Just like how our prior affects the conclusion each person in our group of friends reaches after seeing someone correctly guesses a secret number one hundred times, our choice of the mean function and the kernel determines the predictions made by a GP.

Figure 3.15 shows three examples in which each combination of a mean function and a kernel leads to a drastically different behavior.

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